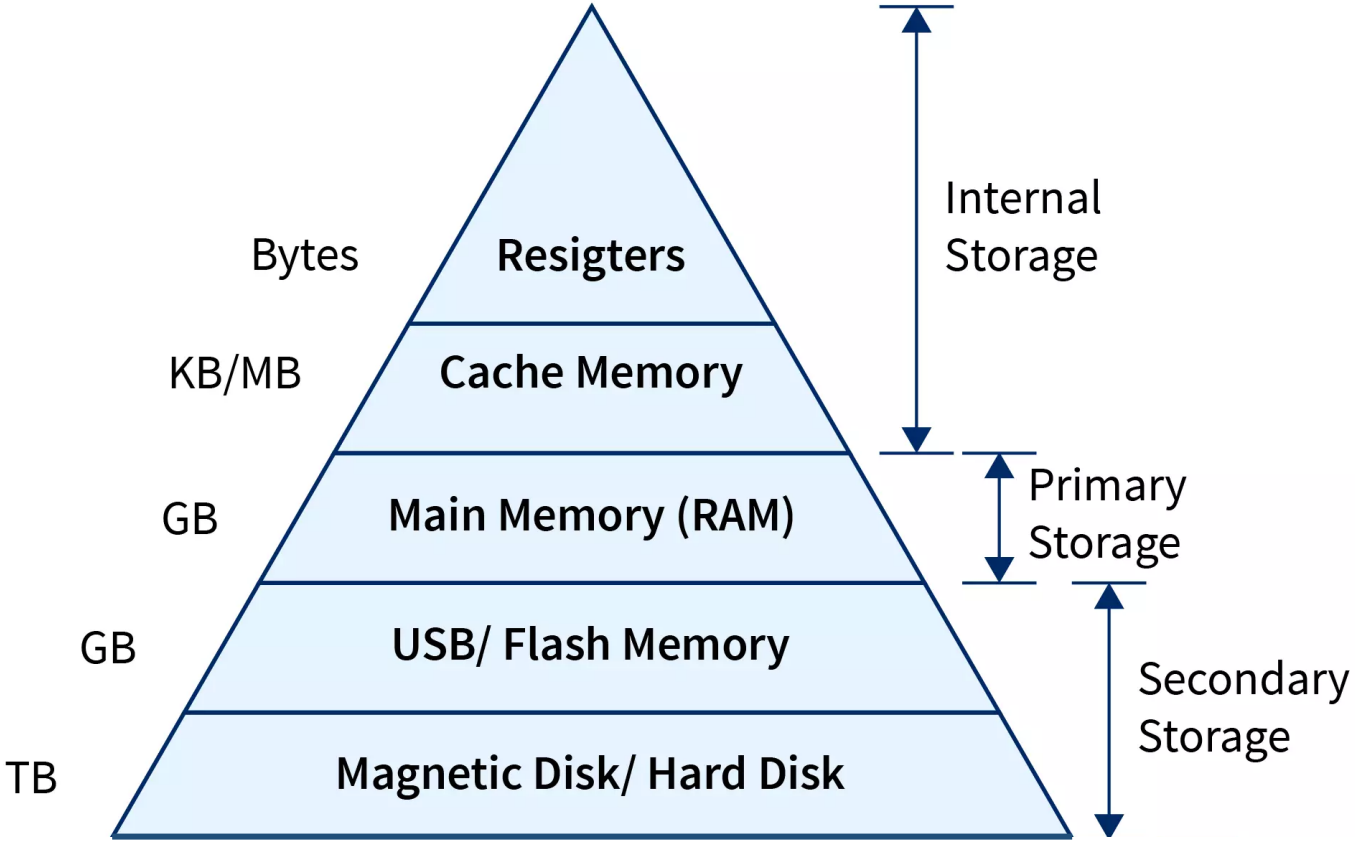
Cache Mapping

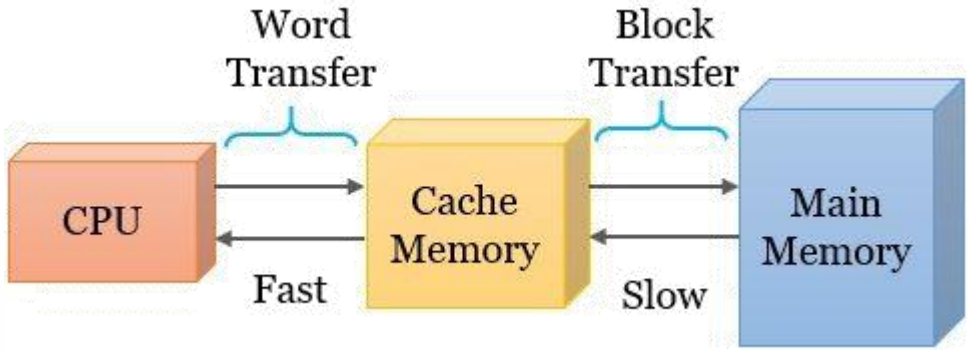
CSE Sem-4

Memory Hierarchy



- **Capacity** increases from register to hard disk
- **Speed** decreases from register to hard disk
- **Cost** increases from register to hard disk

- Cache memory is fast static SRAM
- Dynamic memory is DRAM
- Cache lies between CPU & main memory



- CPU searches word in cache memory first.
- If found – **Cache Hit**
- If not found – **Cache Miss**, searched in main memory
- If found in main memory, page is mapped from main memory to cache memory – **Cache Mapping Techniques**

Cache Mapping Techniques

Parameters of Memory Hierarchy:

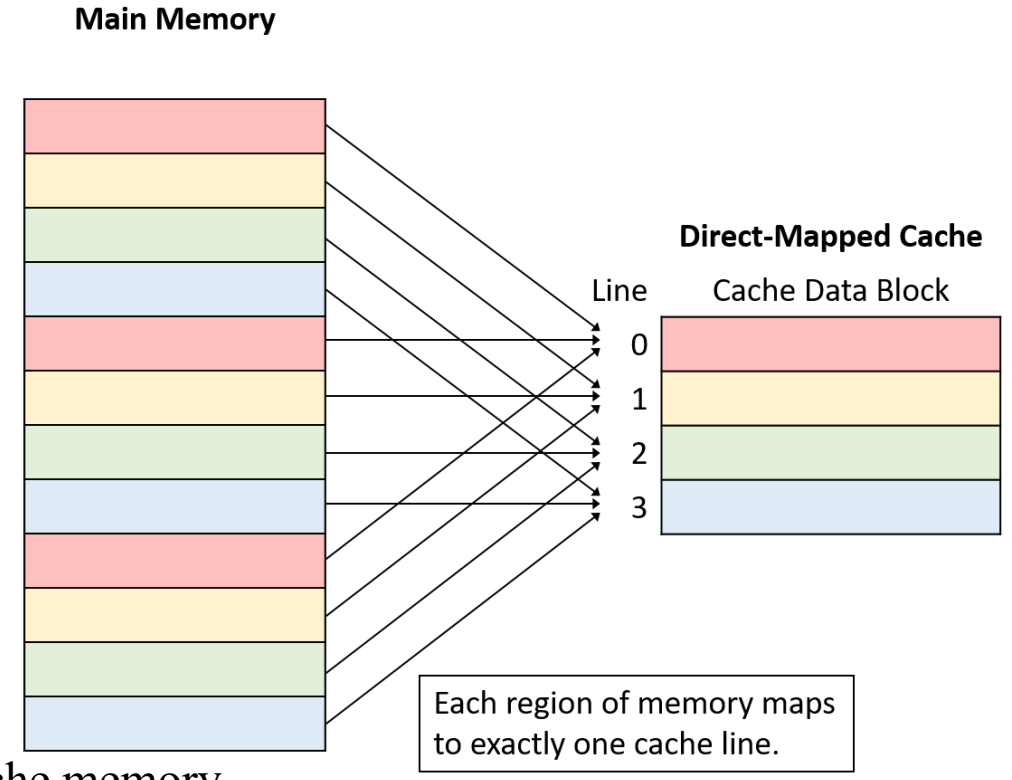
1. Access Time: Time required by the CPU to access memory.
2. Memory Size: Number of bytes in memory.
3. Bandwidth: Rate at which information is transferred between adjacent levels.

- Main memory contain multiple blocks, each block contains a number of words.
- Cache memory contains number of lines.
- Block size of main memory and line size of cache memory is same.
- Allocation of blocks into lines is done using **Cache Mapping Techniques:**
    1. **Direct Mapping**
    2. **Fully Associative Mapping**
    3. **K-way Set Associative Mapping.**
- **Logical addresses** are generated by CPU during execution
- **Physical address** refers to location in a physical memory unit (the one that is loaded into main memory).

Direct Mapping Technique

- Main memory size = 256 words
- Divided into blocks each contain 4 words
- Total no of blocks = 256/4 = 64

- Cache size = 16 words
- Divided into number of lines
- Block size in main memory = line size in cache memory
- Total no of lines = 16/4 =4



Rule for Direct Mapping: k mod n

- k = block number n = number of lines

- B0 (Block 0 of main memory) will be mapped in (0 mod 4) line 0 of cache memory.
- B1 (Block 1 of main memory) will be mapped in (1 mod 4) line 1 of cache memory
- B2 (Block 2 of main memory) will be mapped in (2 mod 4) line 2 of cache memory
- B3 (Block 3 of main memory) will be mapped in (3 mod 4) line 3 of cache memory
- B4 (Block 4 of main memory) will be mapped in (4 mod 4) line 0 of cache memory
- B5 (Block 5 of main memory) will be mapped in (5 mod 4) line 1 of cache memory

- B62 (Block 62 of main memory) will be mapped in (62 mod 4) line 2 of cache memory
- B63 (Block 63 of main memory) will be mapped in (63 mod 4) line 3 of cache memory

Direct Mapping Technique

- A block of main memory means all the 4 words of that block fits into the cache.
- Depending upon the main memory size Physical address size can be generated. So, for main memory of 256 words (256=2^8), 8 bits are required to represents physical address.
- Physical address is divided into two parts – Block number and Block offset.
- Block offset or block size means how many words can be accommodate in each block.
- Size of each block is 4 words. So (4 = 2^2) 2 bits required to represent block offset.
- Total number of blocks 64. So (64 = 2^6) 6 bits required to represent Block number.

❖ For cache memory referencing, cache address is used.

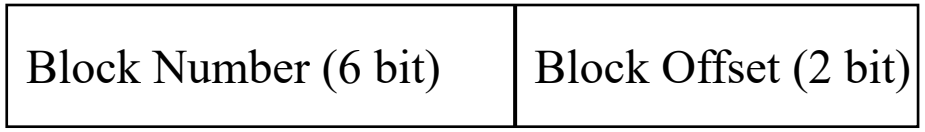
❖ Cache address size and Physical address size are same.

❖ Block offset of cache = Block offset of Main memory.

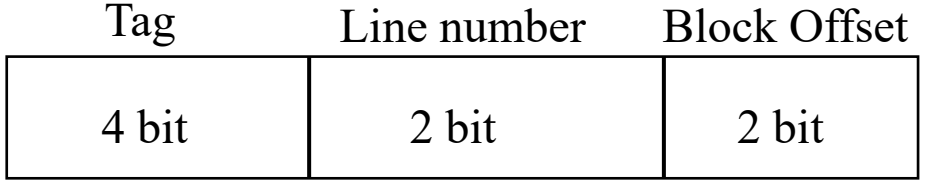
❖ For 4 lines in cache (4 = 2^2) 2 bits required.

❖ Tag is used to identify the block mapped onto one particular cache line. For 64 blocks of main memory, and 4 lines of cache memory, each line can contain (64/4 =) 16 blocks.

❖ For representing 16 blocks, (16= 2^4) 4 bits required to identify the blocks.



8 bits Physical address



8 bits cache address

Fully Associative Mapping Technique

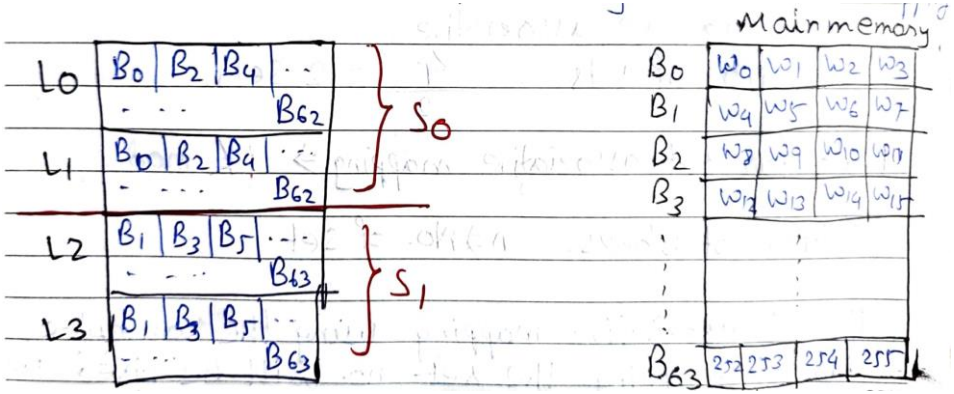
- Each block from main memory, can be mapped into each line of cache memory.
- As any block can be fit into any line of cache memory, there is no line number in address field.
- Any one of the 64 blocks can be fit into each line. So, for tag field (64 = 2^6) 6 bits are required.
- So, tag in cache address = block number in physical address.
- As block size = line size, each block can be mapped at each line, but one at a time.



8 bits Cache address

k-way Set Associative Mapping Technique

- Combination of direct and fully associative mapping
- Cache lines are grouped into sets, where each set contains k number of lines (k-way set associative)
- A particular block can be mapped at a particular set of cache.
- Within that set, block can be mapped to any freely available cache line.



- For a 2-way set associative, 2 lines will be grouped together.
- No of sets for k-way set associative = Total no of lines / k
- No of sets for 2-way set associative = Total no of lines / 2 = 4/2 = 2 (set 0, set 1)
- **Rule for set associative mapping:** M mod n, where M = block number, n = number of sets

- For 2 sets, (2 = 2^1), 1 bit required to represent
- As total number of blocks in main memory is 64, and its divided into two sets, each set contain (64/2 =) 32 blocks. So to represent 32 blocks (32 = 2^5) 5 bits are required for tag field.

Tag	Set number	Block Offset
5 bit	1 bit	2 bit

8 bits cache address

Problems

1. Consider a direct mapped cache of size 32 KB, with block size 32 bytes. The CPU generates 32 bit address. How many bits are required for cache indexing and tag bits?

2. A computer has 512 KB cache memory and 2MB main memory. If the block size is 64 bytes, find the bits required for block number, block size, line number and tag field.

3. Consider a fully associative mapped cache of size 16 KB, with b lock size 256 bytes. The size of main memory is 128 KB. Find the number of tag bits and tag directory size.

4. A 4-way set associative cache memory with a capacity of 16 KB is built using a block size of 8 words. Each word is of 32 bits. The size of physical address space is 4 GB. What will be the number of bits for tag field.